RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Go, change the world

DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester)

Date	**11-12-2024**	**Time**	**10:00 AM-11:30 AM**	
Test	**Test-II**	**Maximum Marks**	**50**	
Course Title	**Fundamentals of Linear Algebra, Calculus & Differential Equations**		**Course Code**	**MA211TB**
Semester	**I**	**Programs**	**AS, CH, IM and ME**	

Sl. No.	**Scheme and Solutions**	M
1(a)	**Show that the curves $r^n = a^n\cos(n\theta)$ and $r^n = b^n\sin(n\theta)$ intersect each other orthogonally.** For the given curves: $r^n = a^n\cos(n\theta), \frac{dr}{d\theta} = -rtan(n\theta), \tan(\phi_1) = r\frac{d\theta}{dr} = -\cot(n\theta)$ and $r^n = b^n\sin(n\theta), \frac{dr}{d\theta} = cot(n\theta), \tan(\phi_2) = r\frac{d\theta}{dr} = tan(n\theta)$ Hence, $\tan(\phi_1)\tan(\phi_2) = -\cot(n\theta)\tan(n\theta) = -1$, curves intersect orthogonally.	2 2 1
1(b)	**Find the slope of the tangent to the cardioid $r = a(1 + cos\theta)$ at the point $\theta = \frac{\pi}{3}$.** Consider $r = a(1 + cos\theta), \frac{dr}{d\theta} = -asin\theta, \tan(\phi) = r\frac{d\theta}{dr} = -\frac{a(1+cos\theta)}{sin\theta} = -\cot\left(\frac{\theta}{2}\right)$ $\tan(\phi) = \tan\left(\frac{\pi}{2} + \frac{\theta}{2}\right), \phi = \frac{\pi}{2} + \frac{\theta}{2}, \psi = \theta + \phi = \frac{\pi}{2} + \frac{3\theta}{2}$, at the point $\theta = \frac{\pi}{3}$ $\text{slope} = \tan(\psi) = \tan(\pi) = 0$	1 2 1 1
2(a)	**The curve that an idealized hanging chain or cable assumes under its own weight when supported only at its ends in a uniform gravitational field is a 'catenary' given by $y = c\cosh(\frac{x}{c})$, show that the curvature at any point on this curve varies inversely as square of ordinate at that point.** $y = c\cosh(\frac{x}{c}), y_1 = \sinh(\frac{x}{c})$ and $y_2 = (\frac{1}{c})\cosh(\frac{x}{c}), \kappa = \left(\frac{1}{c}\right)\cosh\left(\frac{x}{c}\right)/[1 + \sinh^2(\frac{x}{c})]^{3/2}$ $\kappa = \left(\frac{1}{c}\right)\frac{1}{\cosh^2\left(\frac{x}{c}\right)} = \frac{c}{y^2}$, hence $\kappa \propto \frac{1}{y^2}$	1 1 1 1+1
2(b)	**The curve traced by a point on a circle as it rolls along a straight line without slipping is a cycloid $x = a(t - \sin t), y = a(1 - \cos t)$, show that the radius of curvature at any point is $-4a\sin(\frac{t}{2})$** $x' = a(1 - \cos t), y' = a\sin t, x'' = a\sin t, y'' = a\cos t$ $\rho = \frac{[a^2(1 - cost)^2 + a^2\sin^2 t]^{3/2}}{a^2(1 - cost)cost - a^2\sin^2 t} = -4\,\text{asin}\left(\frac{t}{2}\right)$ on simplification	1+1 1+1 1
3	**For the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$, obtain the circle of curvature at the point $(\frac{a}{4}, \frac{a}{4})$.** The first and second derivatives of the given curve at the point $(\frac{a}{4}, \frac{a}{4})$ is $y_1 = -1, y_2 = \frac{4}{a}$ The coordinates for Centre of curvature is $(\bar{x}, \bar{y}) = \left(x - \frac{y_1(1+y_1^2)}{y_2}, y + \frac{(1+y_1^2)}{y_2}\right) = (\frac{3a}{4}, \frac{3a}{4})$	2+2 2+2

	The radius of curvature at the given point is $\rho = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2} = a/\sqrt{2}$ Hence, circle of curvature is $\left(x - \frac{3a}{4}\right)^2 + \left(y - \frac{3a}{4}\right)^2 = \frac{a^2}{2}$	1 1
4	Expand $y = \log(1 + e^x)$ as a Maclaurin series up to the terms containing x^4 and hence obtain the series for $\frac{e^x}{1+e^x}$. The first four derivatives at $x = 0$ is $y_1(0) = \frac{1}{2}, y_2(0) = \frac{1}{4}, y_3(0) = 0 \; and \; y_4(0) = -\frac{1}{8}$ Also $y(0) = \log 2$, Maclaurin series is given by $y = \log(1 + e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \cdots$, The series for $\frac{e^x}{1+e^x} = \frac{d}{dx}[\log(1 + e^x)] = \frac{1}{2} + \frac{x}{4} - \frac{x^3}{48} + \cdots$	1 2 2 2 1 1 1
5(a)	**Solve: $y'' + 2y' + y = e^{2x}$** The auxiliary equation is $m^2 + 2m + 1 = 0, (m+1)^2 = 0, m = -1, -1$ $y_c = (A + Bx)e^{-x}, y_p = \frac{e^{2x}}{4+4+1} = \frac{1}{9}e^{2x}$ The general solution is $y = y_c + y_p = (A + Bx)e^{-x} + \frac{1}{9}e^{2x}$	1 2+1 1
5(b)	**The displacement $x = x(t)$ of a mass-spring system satisfies the differential equation $x''(t) + 9x(t) = 0$, with initial conditions $x(0) = 1, x'(0) = 1$. Obtain $x(t)$ at $t > 0$** The auxiliary equation is $m^2 + 9 = 0, m = \pm 3i$ The general solution is $x(t) = Acos(3t) + Bsin(3t)$, also $x'(t) = -3A\sin(3t) + 3B\cos(3t)$, using the initial conditions we get $A = 1, \; B = \frac{1}{3}$	1 1+1 1 1

******